

## Vernier callipers: Pendulum bob

**Aim:** To measure the diameter of a given pendulum bob (sphere) using a vernier callipers.

**Apparatus:** Vernier callipers, a spherical pendulum bob

**Least count:**  $10 \text{ VSD} = 9 \text{ MSD}$ ,  $LC = \text{MSD} / N = 1 \text{ mm} / 10 = 0.1 \text{ mm} = 0.01 \text{ cm}$

**Zero error:**  $-0.03 \text{ cm}$  ( $n = 7$ :  $-(10 - 7) \times 0.01 \text{ cm}$ ); zero correction =  $+0.03 \text{ cm}$

**Observations: External diameter (d)**

| S.No | MSR (cm) | VSR (n) | $n \times LC \text{ (cm)}$ | Observed (cm) | Corrected (cm) |
|------|----------|---------|----------------------------|---------------|----------------|
| 1    | 1.90     | 8       | $8 \times 0.01 = 0.08$     | 1.98          | 2.01           |
| 2    | 1.90     | 6       | $6 \times 0.01 = 0.06$     | 1.96          | 1.99           |
| 3    | 1.90     | 7       | $7 \times 0.01 = 0.07$     | 1.97          | 2.00           |
| 4    | 1.90     | 5       | $5 \times 0.01 = 0.05$     | 1.95          | 1.98           |
| 5    | 1.90     | 5       | $5 \times 0.01 = 0.05$     | 1.95          | 1.98           |

**Calculations: External diameter (d)**

1. **Mean value:**  $\bar{d} = (d_1 + d_2 + \dots + d_n) / n$

$$\bar{d} = (2.01 + 1.99 + 2.00 + 1.98 + 1.98) / 5 = 1.9920 \text{ cm}$$

2. **Absolute error of each reading:**  $|\Delta d_i| = |\bar{d} - d_i|$

$$|\Delta d_1| = 0.0180, |\Delta d_2| = 0.0020, |\Delta d_3| = 0.0080, |\Delta d_4| = 0.0120, |\Delta d_5| = 0.0120 \text{ cm}$$

3. **Mean absolute error:**  $\Delta \bar{d} = \Sigma |\Delta d_i| / n$

$$\Delta \bar{d} = (0.0180 + 0.0020 + 0.0080 + 0.0120 + 0.0120) / 5 = 0.0104 \text{ cm}$$

4. **Relative error:**  $\delta d = \Delta \bar{d} / \bar{d}$

$$\delta d = 0.0104 / 1.9920 = 0.005221$$

5. **Percentage error:**  $\delta d \times 100 \%$

$$0.522 \%$$

6. **Reporting rule:**  $\Delta d = \max(\Delta \bar{d}, LC)$ , rounded to the LC's decimal place;  $\bar{d}$  rounded to the same place

$$\Delta \bar{d} = 0.0104 \geq LC = 0.01, \text{ so } \Delta d = 0.01 \text{ cm}$$

7. **Result:**  $d = (\bar{d} \pm \Delta d) \text{ cm}$

$$d = (1.99 \pm 0.01) \text{ cm}$$

**Volume of sphere**

1. **Formula:**  $V = (4/3)\pi r^3 = (\pi/6)d^3$

$$V = (\pi/6) \times (1.990)^3$$

2. **Value:**  $V$

$$V = 4.126 \text{ cm}^3$$

3. **Relative error (maximum-error rule):**  $\Delta V/V = 3 \Delta d/d$

$$\Delta V/V = 3 \times 0.01000/1.990 = 0.0151$$

4. **Percentage error:**  $(\Delta V/V) \times 100 \%$

$$1.51 \%$$

5. **Absolute error:**  $\Delta V = (\Delta V/V) \times V$

$$\Delta V = 0.0151 \times 4.126 = 0.0622 \text{ cm}^3$$

## Result

- **External diameter (d):**  $d = (1.99 \pm 0.01) \text{ cm}$
- **Volume of sphere: V = (4.13 ± 0.06) cm<sup>3</sup>**

## Precautions

1. The zero error was noted with the jaws closed and the correction applied with its sign (corrected = observed – zero error).
2. The object was held gently between the jaws: they just touched it without pressing.
3. The slider was locked with the locking screw before each reading.
4. The scales were read with the eye directly above the coinciding lines, to avoid parallax.
5. Readings were taken at different positions / orientations of the object and their mean was used.